

YEAR 11 MATHEMATICS

MID-SEMESTER EXAM

QUESTION 1. (19 marks)

(a) Simplify $\frac{10x+15y}{5x+5y}$ [1]

(b) Factorise $6a^2 - 3a - 3$ [2]

(c) Solve $\frac{x}{3} - \frac{x+1}{2} = 5$ [2]

(d) Show on a number line the solution to $|x+4| \leq 5$ [3]

(e) Solve $x^2 - 7x - 4 = 0$ Answer in exact form. [2]

(f) Factorise $3x^3 - 2x^2 - 6x + 4$ [2]

(g) Simplify $\frac{x^2-3x+2}{x-1} \times \frac{x+1}{x^2-x-2}$ [2]

(h) A number when added to twice its square equals 55.
Find the number(s) that satisfy this condition. [3]

(i) Solve $\frac{1}{x+1} - \frac{1}{2x-1} = 0$ [2]

(j) The volume of a square pyramid is given by $V = \frac{1}{3}x^2h$, where x is the base length
and h is the length of the perpendicular height. Find x , correct to two decimal places,
if $V = 53.77 \text{ m}^3$ and $h = 2.37\text{m}$ [2]

QUESTION 2 (26 marks) START A NEW PAGE

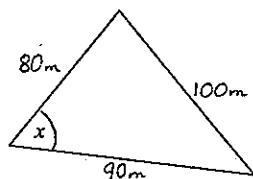
- (a) Write the exact values for:
- (i) $\tan 30^\circ$ [1]
 (ii) $\sec 135^\circ$ [1]

- (b) Find all the values for θ , for $0^\circ \leq \theta \leq 360^\circ$:
- (i) $\sin \theta = 0$ [2]
 (ii) $\cot \theta = \sqrt{3}$ [2]
 (iii) $2 \cos \theta + 1 = 0$ [3]

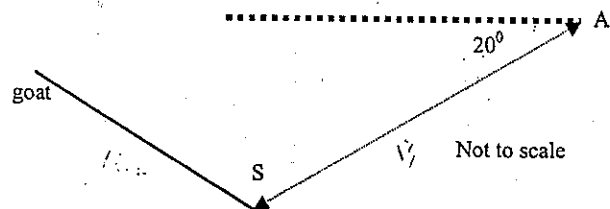
- (c) Simplify:
- (i) $\frac{\sin^2 \theta + \cos^2 \theta}{\tan^2 \theta}$ [2]
 (ii) $1 - \sin^2 (180^\circ + \theta)$ [2]
 (iii) $(\sec^2 \theta - 1) \tan (90^\circ - \theta)$ [3]

- (d) Given that $\tan Y = \frac{2}{3}$, and $\sin Y < 0$, find the exact value of $\sec Y$. [2]

- (e) The sides of a triangular field have lengths 80m, 90m, and 100m.



- (i) Calculate the size of the angle 'x' to the nearest degree. [2]
 (ii) Calculate the area of the triangle. (Answer to 2 significant figures) [2]
 (f) A wooden stake, S, is 13 m from a point, A, on straight fence. SA makes an angle of 20° with the fence. If a goat is tethered to S by a 10m rope, at what angle to the fence is the rope, if the goat reaches the closest possible point on the fence to A. Draw a neat diagram to illustrate your solution.



[4]

QUESTION 3 (19 marks) START A NEW PAGE

- (a) A function $F(x)$ is defined: $F(x) = \begin{cases} x^2 - 2 & \text{when } -3 \leq x \leq 0 \\ 2x + 3 & \text{when } 0 < x \leq 3 \end{cases}$

- (i) Find $(\alpha) F(1)$ [1]
 $(\beta) F(-2)$ [1]

- (ii) Find the range of the function $F(x)$ [1]

- (b) Consider the function $y = -\sqrt{16 - x^2}$

- (i) Write down the domain of the function [1]

- (ii) Write down the range of the function [1]

- (iii) Describe in geometrical terms the number plane graph of $y = -\sqrt{16 - x^2}$ [2]

- (c) Using a clear sketch, show the region determined by the intersection of the following inequalities. Make clear any x and y intercepts.

$$\begin{aligned} y &\geq x^2 - 4 \\ y &\leq 2 \end{aligned} \quad [3]$$

- (d) Sketch neatly the graphs of the following functions. Make clear any x and y intercepts.

- (i) $y = |2x - 1|$ [2]

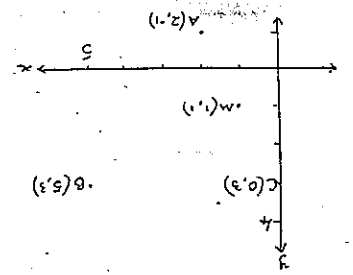
- (ii) $F(x) = \begin{cases} x & \text{when } x > 1 \\ 1 & \text{when } -1 \leq x \leq 1 \\ -x & \text{when } x < -1 \end{cases}$ [3]

- (e) The equation of a circle is $x^2 + y^2 + 4x + 2y - 5 = 0$

- Find (i) The coordinates of the centre of the circle [3]
 (ii) The length of the radius [1]

QUESTION 4 (19 marks) **START A NEW PAGE**

- (a) For the line $2x + 3y - 12 = 0$, find
- (i) the points of intersection with the x and y axes [2]
 - (ii) the gradient of the line [1]
- (b) If $(-1, 2)$ lies on the line $ax - 4y + 11 = 0$, find 'a'. [2]
- (c)
- (i) Plot the points $A(2, -1)$, $B(5, 3)$, and $C(0, 3)$ on a suitably scaled number plane. [2]
 - (ii) Find the midpoint M of AC. [2]
 - (iii) Show that MB is perpendicular to AC. [3]
 - (iv) Find the equation of AC in general form. [3]
 - (v) Find the perpendicular distance from B to AC [2]
 - (vi) Find, using the midpoint formula, the coordinates of D so that ABCD is a parallelogram. [2]



2 6.8
P4
P1

2 6.2
P3

1 6.2
P3

2 6.1
P3

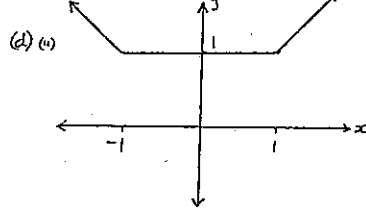
P1
P2
P3

Mark
Cont
Out.

(i) x intercepts @ $y=0$
 $2x - 12 = 0$
 $2x = 12$
 $x = 6 \rightarrow$ accepted
 Ans = (6,0) $\rightarrow$ first answer
 y intercept @ $x=0$
 $3y - 12 = 0$
 $3y = 12$
 $y = 4 \rightarrow$ accepted
 Ans = (0,4) $\rightarrow$ first answer
 (ii) $2x + 3y - 12 = 0$
 $3y = -2x + 12$
 $y = \frac{-2}{3}x + 4$
 $m = -\frac{2}{3}$
 $-a = -3$
 $a = 3$
 $(-1)(-4(2) + 11) = 0$

QUESTION FOUR

QUESTION THREE (cont)



(e) $x^2 + y^2 + 4x + 2y - 5 = 0$
 $(x^2 + 4x + 2^2) + (y^2 + 2y + 1^2) = 5 + 4 + 1$
 $(x+2)^2 + (y+1)^2 = 10$

(i) Centre: (-2, -1)

(ii) Radius: $\sqrt{10}$

Mark
Cont
Out.

3 4.2
P

3 4.3
P

1 4.3
P1

(vi) $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
 $(1,1) = \left(\frac{x_1 + 5}{2}, \frac{y_1 + 3}{2} \right)$
 $x_1 = -3$
 $y_1 = -1$
 Best answer: D (-3,-1)
 Also accepted:
 D (3,7)
 D (7,-1)

(v) $d = \sqrt{a^2 + b^2}$
 $= \sqrt{2(5)^2 + 1(3)^2}$
 $= \sqrt{50 + 9}$
 $= \sqrt{59}$
 $= 2\sqrt{59}$
 Must mention that
 BM $\perp$ AC for
 full marks

(iv) $y - y_1 = m(x - x_1)$
 $y - 3 = -2(x - 0)$
 $y = -2x + 3$
 $2x + y - 3 = 0$

(iii) Grad. MB = $\frac{3-1}{5-1} = \frac{2}{4} = \frac{1}{2}$
 Grad AC = $\frac{-1-3}{2-0} = \frac{-4}{2} = -2$
 $\frac{1}{2} \times -2 = -1$
 $m_1 m_2 = -1$
 $m_1 = -2$
 $y - y_1 = m(x - x_1)$
 $y - 3 = -2(x - 0)$
 $y = -2x + 3$
 $2x + y - 3 = 0$

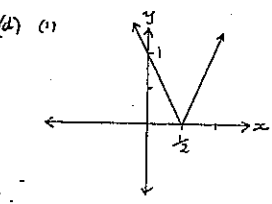
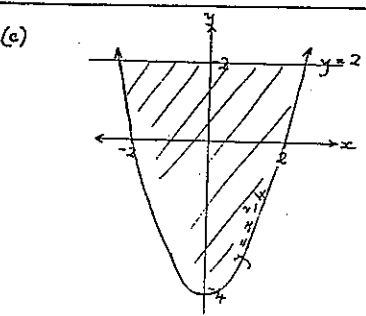
(ii) $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
 $= \left(\frac{2+0}{2}, \frac{-1+3}{2} \right)$
 $= (1,1)$

QUESTION FOUR (cont)

QUESTION THREE

(a) (i) $F(1) = 2(1) + 3 = 5$
 (ii) $F(-2) = (-2)^2 - 2 = 2$
 (iii) $-2 \leq y \leq 9$

(b) (i) $-4 \leq x \leq 4$
 (ii) $-4 \leq y \leq 0$
 (iii) Semi-circle, concave up
 centre (0,0), radius 4 units



Mark
Cont
Out.

1 4.1
P5

1 4.1
P3

2 4.4
P4

2 4.2
P1

P4
P5

Mark
Cont
Out.

QUESTION FOUR

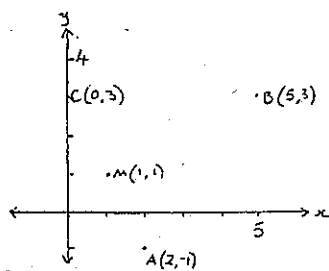
(a) (i) x intercepts @ $y = 0$
 $2x - 12 = 0$
 $2x = 12$
 $x = 6 \leftarrow \text{accepted}$
 Ans = $(6, 0) \leftarrow \text{best answer}$

y intercept @ $x = 0$
 $3y - 12 = 0$
 $3y = 12$
 $y = 4 \leftarrow \text{accepted}$
 Ans = $(0, 4) \leftarrow \text{best answer}$

(*) $2x + 3y - 12 = 0$
 $3y = -2x - 12$
 $y = -\frac{2}{3}x - 4$
 $m = -\frac{2}{3}$

b) $(-1) \cdot -4(2) + 11 = 0$
 $-a = -3$
 $a = 3$

c) (i)



QUESTION FOUR (cont)

(ii) $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
 $= \left(\frac{2+0}{2}, \frac{-1+3}{2} \right)$
 $= (1, 1)$

(iii) Grad. MB = $\frac{3-1}{5-1}$
 $= \frac{1}{2}$

Grad AC = $\frac{-1-3}{2-0}$
 $= -2$

$m_1, m_2 = -1$
 $\frac{1}{2}x - 2 = -1$

(iv) $y - y_1 = m(x - x_1)$
 $y - 3 = -2(x - 0)$
 $y = -2x + 3$
 $2x + y - 3 = 0$

(v) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$
 $= \frac{|2(5) + 1(3) - 3|}{\sqrt{4 + 1}}$
 $= \frac{10}{\sqrt{5}} = 2\sqrt{5}$

In future,
 Must mention that
 BM $\perp$ AC for
 full marks

(vi) $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
 $(1, 1) = \left(\frac{x+5}{2}, \frac{y+3}{2} \right)$
 $x = -3$
 $y = -1$

Best answer: D(-3, -1)
 Also accepted:
 D(7, -1)
 D(3, 7)